Field Data Quality Monitoring Project

- Data: Survey data stimulated modelled on SurveyCTO export format.

Column Names	
submission_id	safety_perception
enumerator_id	bystander_comfort
route_id	harassment_exp
interview_date	harassment_type
duration_min	gps_lat
respondent_gender	gps_lon
respondent_age	

- Tools used: R
  - **Dplyr:** for data manipulation
  - **Tidyr:** for data manipulation, specifically to enable use of pivot_longer(), to convert enumerator summary from a wide to a long table to create the bar chart using ggplot2.
  - **Ggplot2:** Data visualization
  - **Lubridate:** to create the date sequence for the stimulated SurveyCTO raw data
  - **Forcats:** for factor reordering when creating the P1 chart, so that enumerator ID is ordered from lowest to highest flag percentage instead of alphabetical order.

This project illustrates a data quality monitoring workflow using a sample dataset generated in the format of a SurveyCTO field export on R. The dataset used in this project is modelled on a hypothetical passenger survey for a study on gender-based harassment in public transit, and the data fields are listed in the table above.
Two types of checks are done in this project, supported by visualisation to aid next steps.

The first is a data quality check, which includes the following:

1. Checking for missing values per variable and calculating the percentage
2. Anomalies in interview duration to flag submissions that are too long or too short (threshold is usually set based on average duration per survey calculated from the pilot)
3. Internal consistency errors between conditional survey items.
4. Checking GPS coordinates to see if survey location is within the expected radius of the field locations.

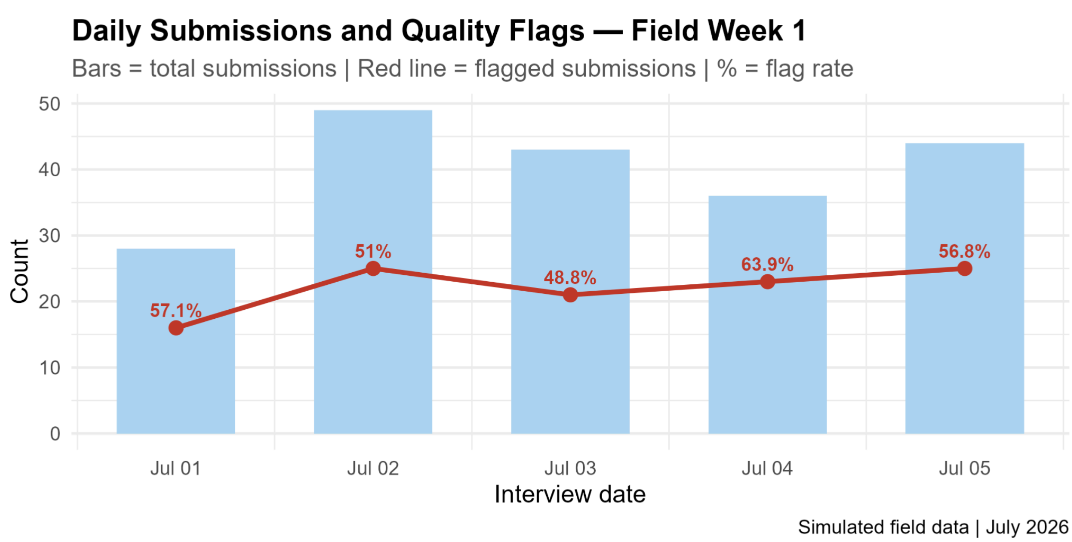
The second is an enumerator performance summary which includes the following:

1. Duration check: Number of submissions from each enumerator ID that are too short or too long, in relation to the average duration calculated for that question from the pilot (or previous waves/days of the study)
2. Consistency check: Number of submissions from each enumerator ID that are logically inconsistent. For example, if a respondent reports harassment experience as “Yes”, but reports harassment type as “None”, this is marked as an inconsistency flag.
3. Location check: Number of submissions from each enumerator ID that are beyond the expected radius of the field location.

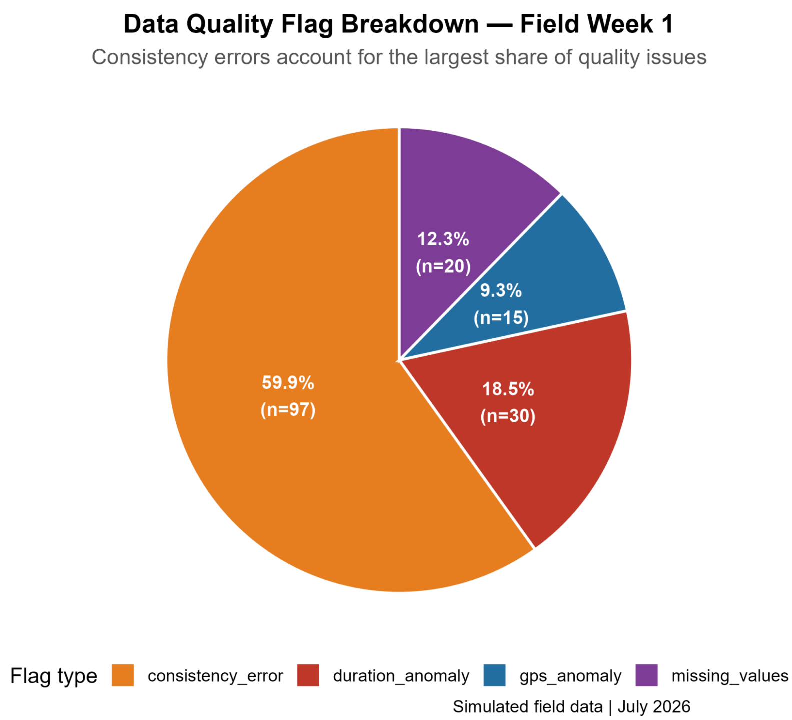
Data quality checks are the first step in diagnosing a problem with the survey and can point towards various issues. Missing values and inconsistent entries can indicate that respondents find the survey question difficult to answer or that the question is unclear/confusing. It could also indicate the need to mark certain questions as mandatory on the survey form, if an important data point is being left out. Anomalies in interview duration and GPS flags can indicate issues with enumerator performance. Survey durations that are too short or too long, and GPS coordinates that fall beyond the field radius can point towards fraudulent entries being made. Survey duration can also indicate whether enumerators need a refresher training to understand the questionnaire better and conduct the survey appropriately.

Enumerator performance summary is useful especially when data quality is found to be compromised due to enumerator performance. Summarising the duration flags, consistency flags and location flags per enumerator ID will help identify individuals who refresher training, and also help track repeated flags that need attention or escalation.

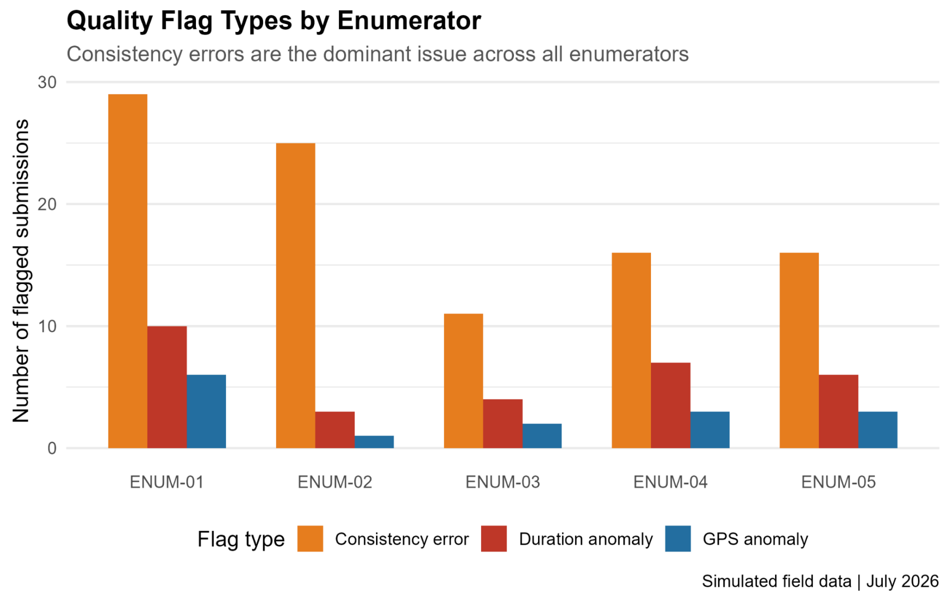
The charts prepared from the sample dataset are as follows:

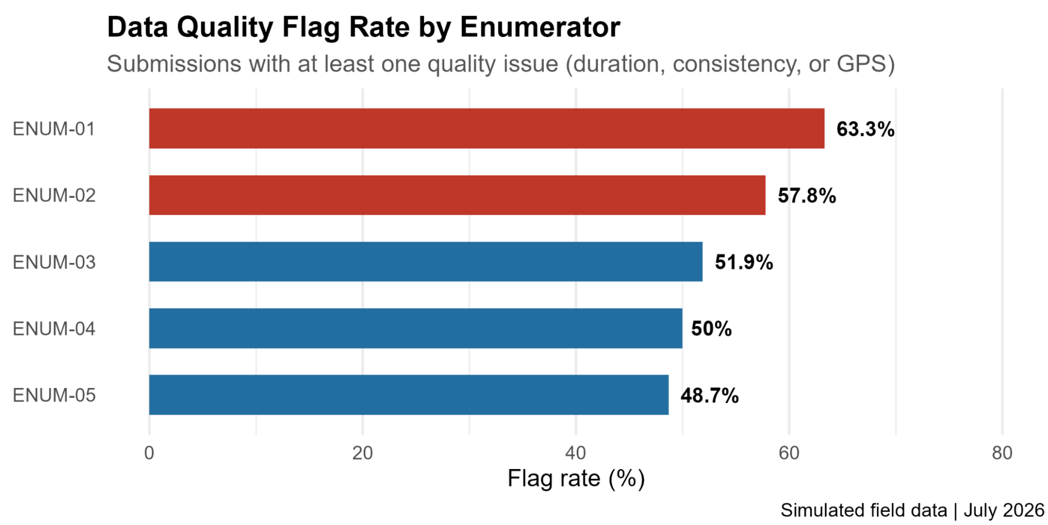


This chart illustrates the total number of submissions per day during the first week of the survey. The line graph shows the percentage of submissions out of the total that have some type of issue flagged by the data quality check, each day. A spike in daily flag rates may also point towards an instrument issue or a field condition that affected all enumerators on that day. If more than 50% of submissions completed each day have some kind of data quality issue, this is a cause for concern. Knowing the type of data quality issue is important to determine next steps.



The second chart shows us the percentage breakdown of the types of data quality flags in the survey data from the first week. Inconsistency errors seem to be the highest in this situation. Hence, the logical plan of action would be to examine the questions and evaluate if the issue arose out of a problem in the questionnaire design.





If the questionnaire is not the source of the issue, the two charts above will help us identify if the issue is related to enumerator performance. Data inconsistency is high among all 5 enumerators listed on the chart, which may indicate the need for a refresher training or clarifying to the enumerators on how and what exactly needs to be recorded for each question on the survey. The last chart helps us identify the enumerators who have the most data quality flags in their submissions.